

Interoperability Conformance Specification: hybridMultiSpectralEncoding

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Foreword

This document has been prepared following the [ICC Intellectual Property Policy](#). This policy is based on the ITU-T/ITU-R/ISO/IEC [Guidelines for Implementation of the Common Patent Policy](#) (23 April 2012), with [interpretations and clarifications](#) to make it specific to ICC. A [Patent Statement and Licensing Declaration form](#) is available.

ICC Interoperability Conformance Specifications, of which this document is an example, may be submitted to the competent ISO Technical Committee for consideration and development as an ISO document. If so, this foreword is to be replaced by the appropriate wording supplied by ISO.

Introduction

ISO 20677-1 defines specifications that provide a platform for defining extended (iccMAX) colour management profiles and systems for various colour workflow domains. It provides a platform for which domain specific specifications can be defined that make use of iccMAX extensions to the existing cross-platform profile format of ISO 15076-1 (ICC) with greater flexibility for defining colour transforms and profile connection spaces to meet needs outside the scope of ICC profiles. However, not all systems provide support for iccMAX profiles thus limiting the ability to take advantage of extended capabilities at strategic points in the color management ecosystem. Using a hybrid ICC profile that has an embedded iccMAX profile as a tag allows for compatibility with ICC systems that do not support iccMAX functionality while providing systems in the color management ecosystem with iccMAX extensions to take advantage of extended color management capabilities enabled by iccMAX profiles.

Requirements specifying restrictions to iccMAX that apply to a particular workflow are defined in workflow domain specifications known as Interoperability Conformance Specifications, of which this document is one example. Additionally, for some domain specific workflows it is envisioned that workflows will connect both to ICC profiles, iccMAX profiles, and hybrid profiles containing both ICC and iccMAX content.

An Interoperability Conformance Specification (ICS) is approved and registered by the International Color Consortium (ICC). It defines minimum structural and operational requirements for writing and reading iccMAX profiles in order to address a specific problem and/or functionality that cannot readily be handled using the ICC profile format defined by ISO 15076-1. An ICS document essentially defines restrictions to iccMAX content in profiles for specific use cases.

Interoperability Conformance Specification: hybridMultiSpectralEncoding

1 Scope

This specification defines workflow requirements and restrictions for hybrid ICC profiles that contain an iccMAX sub-profile as a tag for the purpose of providing connection between an x -channel multi-spectral colour encoding space and a spectral reflectance Profile Connection Space (PCS), where x represents the number of channels in the multi-spectral colour encoding space. The multi-spectral color encoding space has two parts. The first channels of the encoding space have a direct relationship to colorimetry. The rest of the channels provide additional multi-spectral channels for conversion to/from a spectral reflectance PCS. The base part of the hybrid profile provides colorimetric PCS access for systems that do not support iccMAX while the iccMAX sub-profile provides spectral reflectance PCS access. This allows for multi-spectral images to be created and exchanged that have some level of backwards compatibility with systems that do not support spectral image processing.

The particular sub-set of the tags in the iccMAX sub-profile defined in ISO 20677-1 that are required to be present is defined, together with any optional tags that are permitted. The connections between profiles are described, and the processing elements that the CMM is required to support are identified.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 15076-1:20##, *Image technology colour management — Architecture, profile format, and data structure*

NOTE: The most recent version of the ICC specification is available on the ICC web site [2].

ISO 20677-1:20##, *Image technology colour management — Extensions to architecture, profile format, and data structure: iccMAX*

NOTE: The most recent version of the iccMAX specification is available on the ICC web site [2].

3 Terms and definitions

All terms and definitions relevant to this document are provided in ISO 15076-1 and ISO 20677.

4 Use case

4.1 Domain

Profiles and workflows conforming to this ICS shall apply to the domain of colour space conversions including acquisition, preview and output of spectral data.

4.2 Intended use

This specification defines workflows in which colour data is converted either to or from spectral reflectance. The base ICC profile portion of a hybrid multi-spectral encoding profile (defined by ISO 15076-1) provides transforms between a subset of the multi-spectral encoding space and a colorimetric representation. The iccMAX portion of a hybrid multi-spectral encoding profile (defined by ISO 20677-1 and this ICS) provides transforms between the full multi-spectral color encoding space and a spectral reflectance PCS (where subsequent conversion to colorimetry can be achieved by applying the illuminant and observer specified in the spectralViewingConditionsTag).

When used in a workflow in which the CMM handling colour transforms is unable to process the iccMAX portion of the hybrid profile, the base ICC profile portion is used, the iccMAX portion ignored, and the conversion is from the subset of the multi-spectral colour space to colorimetric PCS. Otherwise, the CMM converts between the multi-spectral encoding space and spectral channels, with the option to apply further transforms depending on the workflow configuration and connection scenario. Thus, the hybridMultiSpectralEncoding profile can be used when it is desired to convert between a multi-spectral encoding space, spectral reflectance, and/or non-standard colorimetry.

4.3 Restrictions

ISO 15076-1 provides full details for an ICC *colorspace* class profile. This document defines a set of restrictions which apply to an iccMAX sub-profile when it is embedded into an ICC *colorspace* profile, as a component of a hybrid profile created for the specific use cases described above. Such restrictions include the sub-set of tags from ISO 20677-1 which are permitted in such an iccMAX sub-profile conforming to this document.

5 Workflow

5.1 Hybrid sub-profile sub-classes

The profile class of a hybrid iccMAX sub-profile shall match the class of the containing ICC profile.

A supporting CMM shall support the profile classes and sub-profile sub-classes identified in Table 1 for conformance with this ICS. All profile classes are defined in ISO 15076-1 and ISO 20677-1.

Table 1. Profile classes and subclasses defined by this ICS

Profile	Profile class	Sub-profile sub-class identifier	Class signature	Sub-class signature
	colorspace	multiSpectralEncoding	'spac'	'mspc'

5.2 Connection Scenarios

5.2.1 General

The ICC profile portion of a hybrid colorspace profile is intended to be used as either a source profile or a destination profile by any system that supports ICC colorspace class profiles.

The ‘mspc’ multiSpectralEncoding sub-profile in a hybrid multi-spectral encoding profile connects x input channels (where x corresponds to the number of channels of the Data colour space field of the header) to either a colorimetric Profile Connection Space or to a spectral Profile Connection Space containing n spectral channels (where n corresponds to the number of spectral PCS channels defined in the header). It can be used as either a source profile, or a spectral reproduction target profile, using a conversion between data channels and spectral channels.

5.2.2 Profiles for ICS workflow scenarios

Table 2 provides overview information about additional profiles that are used to describe several profile connection scenarios conforming to this ICS.

Table 2. Additional profiles referenced in workflow scenarios defined by this ICS

Profile	Description
	Profile conforming to ISO 20677-1 or ISO 15076-1 containing (a) colorimetric transform(s) with CMM configured to use a colorimetric transform type. If a separate ICS specification is associated with this profile it should be followed in addition to the connection requirements of this document.
	Profile conforming to ISO 20677-1 containing (a) spectral reflectance transform(s) with the CMM configured to use a spectral transform type. If a separate ICS specification is associated with this profile it should be followed in addition to the connection requirements of this document.
	Profiles conforming to ISO 20677-1 containing Profile Connection Condition tags (spectralViewingConditionsTag, customToStandardPccTag, standardToCustomPccTag)

5.2.3 Workflow Scenarios

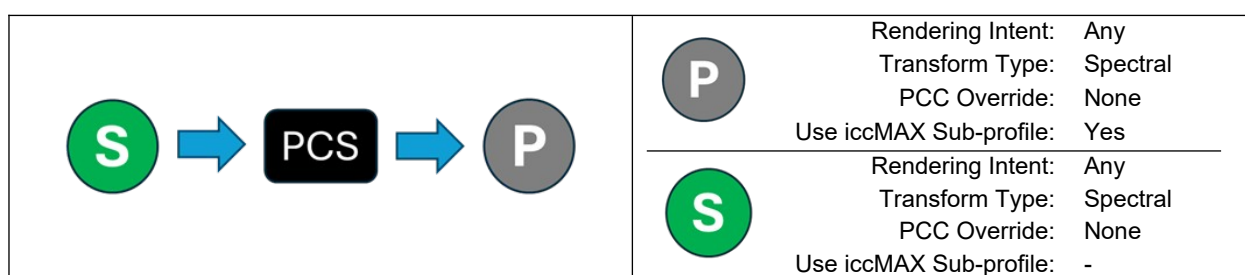
5.2.3.1 General

The following sections define scenarios that conform to this ICS.

Note: systems that only support ICC profiles will only work with scenarios 5.2.3.2 and 5.2.3.3.

5.2.3.2 Scenario 1: Connecting a source spectral profile to a destination hybrid multi-spectral encoding profile and hybrid multi-spectral image with a spectral reflectance PCS (multi-spectral color space encoding)

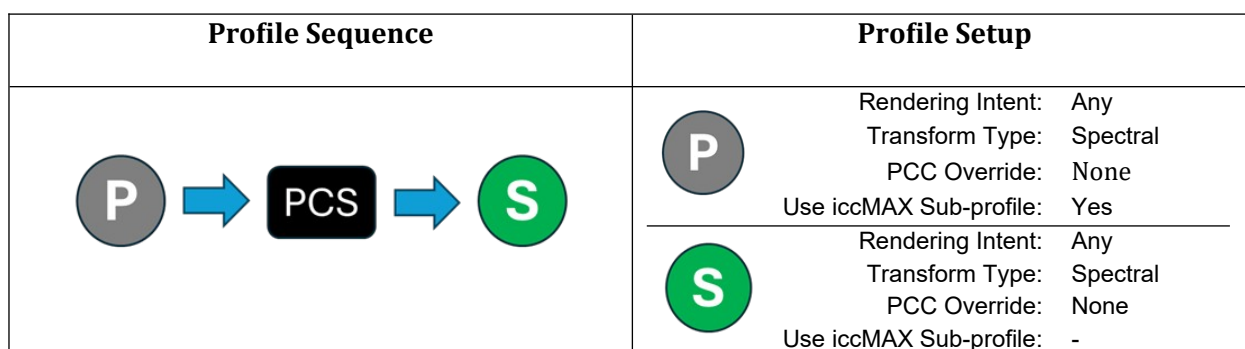
Profile Sequence	Profile Setup
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This scenario may be referred to as a spectral “encoding” workflow as it uses the iccMAX sub-profile transforms in the hybrid colorspace profile convert to x-channel data from spectral reflectance PCS data that come from the source profile. Spectral PCS operations may be utilized as outlined in ISO 20677-1. Systems that do not support iccMAX profiles do not support this scenario.

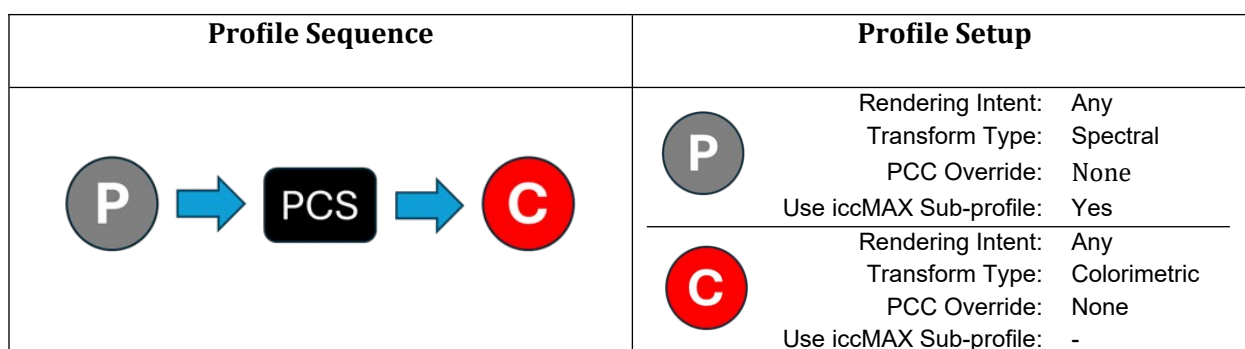
Note: The application of profile P results in extra channels that can be saved as a resulting hybrid image with profile P embedded. In this case, the resulting image can be used by systems that do not support iccMAX profiles as the base part of the profile will work with its associated image channels (see 5.2.3.6 scenario 6) ignoring the extra channels.

5.2.3.3 Scenario 2: Connecting a hybrid multi-spectral encoding profile and hybrid multi-spectral image as source to a destination profile with a spectral reflectance PCS (multi-spectral color space decoding)



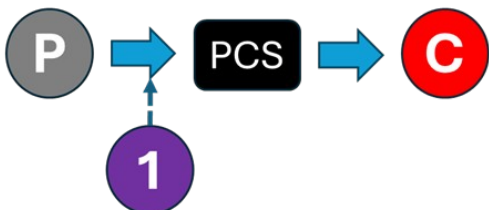
This scenario may be referred to as a spectral “decoding”, sampling” or “conversion” workflow. It uses the iccMAX sub-profile transforms in the hybrid colorspace profile convert x-channel data of a hybrid image to spectral reflectance, which is then passed to the destination profile. Spectral PCS operations may be utilized as outlined in ISO 20677-1. Systems that do not support iccMAX profiles do not support this workflow.

5.2.3.4 Scenario 3: Connecting a hybrid multi-spectral encoding profile and hybrid multi-spectral image as source to a destination profile with a colorimetric PCS (multi-spectral preview)



This scenario converts *x*-channel data spectral encoding data to colorimetry (that can be used as a colorimetric preview) , It uses the iccMAX sub-profile transforms in the hybrid colorspace profile to convert source data to colorimetry for display or output. In an *x*-channel hybrid image intended for use in this workflow, the first channels are used by the base profile to get colorimetric PCS data, and the extra channels provide additional information used for spectral encoding. In this case the iccMAX sub-profile transform results in spectral reflectance PCS data. This spectral reflectance PCS data is then converted to colorimetry in conjunction with Profile Connection Condition. The resulting PCS data is then passed to the destination profile C transform. Systems that do not support iccMAX profiles do not support this workflow.

5.2.3.5 Scenario 4: Connecting a hybrid multi-spectral encoding profile as source using a PCC override to a destination profile with an alternate colorimetric PCS (multi-spectral preview with alternate observing conditions)

Profile Sequence	Profile Setup				
	<table> <tr> <td></td><td> Rendering Intent: Any Transform Type: Spectral PCC Override: 1 Use iccMAX Sub-profile: Yes </td></tr> <tr> <td></td><td> Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: - </td></tr> </table>		Rendering Intent: Any Transform Type: Spectral PCC Override: 1 Use iccMAX Sub-profile: Yes		Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: -
	Rendering Intent: Any Transform Type: Spectral PCC Override: 1 Use iccMAX Sub-profile: Yes				
	Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: -				

This scenario is nearly the same as the previous scenario except that alternate Profile Connection Conditions provided by Profile 1 are used to convert the spectral PCS data from the transform of source profile P to colorimetry that will be passed into the transform of destination profile C. Systems that do not support iccMAX profiles do not support this workflow.

5.2.3.6 Scenario 5: Connecting a hybrid multi-spectral encoding profile as source to a destination profile with a colorimetric PCS (legacy multi-spectral preview)

Profile Sequence	Profile Setup				
	<table> <tr> <td></td><td> Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: No </td></tr> <tr> <td></td><td> Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: - </td></tr> </table>		Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: No		Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: -
	Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: No				
	Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: -				

This scenario uses the transforms in the encapsulating ICC profile to convert channels associated with the base of profile P to colorimetry for display or output. Additional channels (if they exist in the source image are ignored). In this scenario the iccMAX sub-profile is not used for profile P, as the base ICC profile is used instead. This is the default behavior of systems that do not support iccMAX profiles when this profile is used as a source profile.

5.2.3.7 Scenario 6: Connecting a source profile to a hybrid multi-spectral profile as destination with a colorimetric PCS (legacy to partial multi-spectral conversion)

Profile Sequence	Profile Setup				
	<table> <tr> <td></td><td> Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: - </td></tr> <tr> <td></td><td> Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: No </td></tr> </table>		Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: -		Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: No
	Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: -				
	Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: No				

This scenario converts from colorimetry coming out of the transform in source profile C to device encoding using the output transform of the base part of destination profile P. Additional spectral encoding channels will not be part of the output of this scenario - as the base part of profile P only encodes a subset of the channels that the iccMAX sub-profile encodes.

Caution: This is the default behaviour of systems that do not support iccMAX profiles when this profile is used as a destination profile. The IccMAX sub-profile will not be usable if only the base subset channels are stored in an image with an embedded profile P. Therefore, this scenario is probably one that should be avoided.

6 Sub-class Profile Requirements

6.1 General

Requirements for hybrid iccMAX profiles conforming to this ICS document are listed here. ICC v4 profiles shown in Section 5 above shall conform to ISO 15076-1.

6.2 Requirements

The requirements for a hybrid multi-spectral encoding profile are in two parts.

The encoding of the base profile shall be as defined in ISO 15076-1 as a colorspace class profile.

An extension iccMAX sub-profile shall also be encoded in this base (encapsulating) profile as defined in the embedding annex in ISO 20677-1. This involves adding an embeddediccMAXProfileTag (signature 'ICC5') to the base profile, with the tag encoded as an embeddedProfileType (signature 'ICCP') with the sub-profile encoded as defined in ISO 20677-1.

The encoding of the profile header of the encapsulated sub-profile in the embeddediccMAXProfileTag shall be as defined in ISO 20677-1, with the specific requirements shown in Table .

Full details of the encoding of the header fields in 3 are given in ISO 20677-1.

Table 3. Header requirements

Header field	Required content
Profile class	'spac'
Profile subclass	'mspc'
Profile subclass major version	1
Profile subclass minor version	0
Profile flags	Embedded
Device attributes	
Data colour space	'xCLR' where <i>x</i> refers to number of Data colour channels. The color space shall be the same as in the encapsulating ICC profile.
MCS	0
Colorimetric PCS	0
Spectral PCS	'rsxxxx' where xxxx is a hex value corresponding to the number of spectral channels in the PCS
Spectral PCS range	Start and end wavelengths of the spectral PCS encoded as float16 values
Bispectral PCS	0

6.2.1 Required tags

The encapsulated sub-profile shall contain the tags listed in Table .

Table 4. Required tags

Tag name	Signature	Required content
spectralWhitePointTag	'swpt'	float16NumberType containing an array of values defining the spectral reflectance or emission of the media white point
BToD3	'B2D3'	multiProcessElementType element containing zero or more curveSetElement, matrixElement, CLUTEElement, extendedCLutElment processing elements
DToB3	'D2B3'	LutAToBType or multiProcessElementType element containing zero or more curveSetElement, matrixElement, CLUTEElement, extendedCLutElment processing elements
spectralViewingConditionsTag	'svcn'	Structure defining Standard 2-degree observer, and D50 illuminant

The encoding of the tags listed in Table 4 shall be as defined in ISO 20677-1.

6.2.2 Example

The profile .icc is encoded according to the requirements of this ICS document and is available in the iccDEV-Testing suite [3]. An XML representation is also provided.

7 Conformance

A profile shall be considered to be in conformance with this ICS document if it meets the following conditions:

- The profile connects to the channels specified in section 5.
- The profile header includes the required content from Table 3.
- All required tags listed in Table 4 are present in the profile.
- The profile structure and all tags conform to ISO 20677-1.

A CMM shall be considered to be in conformance with this ICS if it meets the following conditions:

- The CMM is able to parse profiles that conform to this ICS
- The CMM supports and is capable of processing the channels specified in section 5 and any other profiles listed in Table 1.
- The CMM is able to process the tags listed in Table 4.
- When processing a profile conforming to this ICS, the CMM produces results that are a close approximation to those produced by the iccDEV implementation [3]

Bibliography

[2] iccMAX <http://www.color.org/iccmax/>

[3] iccDEV <http://www.color.org/iccmax/index.xalter#iccdev>

8 Annex A (informative) - Worked example scenarios

This annex describes the worked example scenarios provided in the iccDEV-Testing ICS package named HybridMultiSpectralEncoding. Each entry is driven by a JSON configuration file in the package's config/ directory and demonstrates one of the workflow scenarios defined in clause 5.2.3 applied to the example hybrid multi-spectral encoding profile. The spectral test image used by these examples, Data/smCows380_5_780.tif, is a "metacow" metamerism target: the head and tail halves of each cow have different spectral reflectances that form a metameric pair under CIE Illuminant D65 with the 1931 standard 2-degree observer, but diverge under other illuminants. This is why Scenarios 3 and 4 preview the encoding under several illuminants - under D65 the head and tail halves match, while under Illuminant A, D93 and F11 they differ, a behaviour that only a spectral (rather than single-illuminant colorimetric) encoding can reproduce.

Table A.1. Worked example scenarios in the iccDEV-Testing ICS package

ID	Clause 5.2.3 scenario	iccDEV tool / driver file	What is demonstrated
S1	Scenario 1 (5.2.3.2)	iccApplyProfiles / hmse-S1-refCowsToMsCows.json	Encoding a spectral reflectance image (smCows, 380-780 nm) into the eight-channel multi-spectral encoding via the iccMAX sub-profile of the example profile.
S2	Scenario 2 (5.2.3.3)	iccApplyProfiles / hmse-S2-msCowsToRefCows.json	Decoding the multi-spectral encoding back to a 36-band spectral reflectance image via the iccMAX sub-profile.
S3	Scenario 3 (5.2.3.4)	iccApplyProfiles / hmse-S3-previewMSCowsD50.json	Colorimetric preview of the multi-spectral encoding under its native D50 viewing conditions, using the iccMAX sub-profile.
S4	Scenario 4 (5.2.3.5)	iccApplyProfiles / hmse-S4-previewMSCows{A,D65,D93,F11}.json	Colorimetric preview under alternate observing conditions (CIE Illuminant A, D65, D93 and F11) using a PCC override on the iccMAX sub-profile.
S5	Scenario 5 (5.2.3.6)	iccApplyProfiles /	Legacy preview using

		hmse-S5- previewRgbCowsD50.js on	only the base ICC part of the example profile, with the iccMAX sub- profile ignored.
S6	Scenario 6 (5.2.3.7)	iccApplyProfiles / hmse-S6- rgbToHalfMS.json	Legacy-to-partial conversion: an sRGB image rendered into the base channels only, producing a "half" multi-spectral image whose extra spectral channels are not populated.
[comparison]	[reference]	iccApplyProfiles / hmse- previewRefCows{D50,A ,D65,D93,F11}.json	Reference previews produced directly from the spectral reflectance image under the same five conditions as Scenarios 3 and 4, for comparison against the multi-spectral-encoding previews.
S5 (half MS)	Scenario 5 (5.2.3.6)	iccApplyProfiles / hmse-S5- previewHalfMS.json	Preview of the half multi-spectral image produced by Scenario 6 through the base ICC profile. This succeeds, because only the base channels are required.
S3 (half MS)	Scenario 3 (5.2.3.4)	iccApplyProfiles / hmse-S3- previewHalfMS.json	Attempted spectral preview of the same half multi-spectral image through the iccMAX sub-profile. EXPECTED TO FAIL with a device-channel mismatch: the image has 3 channels but the sub-profile requires 8. Demonstrates the caution in clause 5.2.3.7.

All scenarios in Table A.1 complete successfully except the final entry, which is included deliberately to illustrate a non-conforming use. Running `hmse-S3-previewHalfMS.json` is expected to fail: the `iccMAX` sub-profile of the example profile requires the full eight-channel multi-spectral encoding, but the half multi-spectral image produced by Scenario 6 carries only the three base channels, so the conversion stops with a device-channel mismatch ("Number of samples 3 in image ... doesn't match device samples 8 in first profile"). This is the practical consequence of the caution given in clause 5.2.3.7. The `BuildAndTest.bat` and `BuildAndTest.sh` scripts label this step as an expected failure and continue.